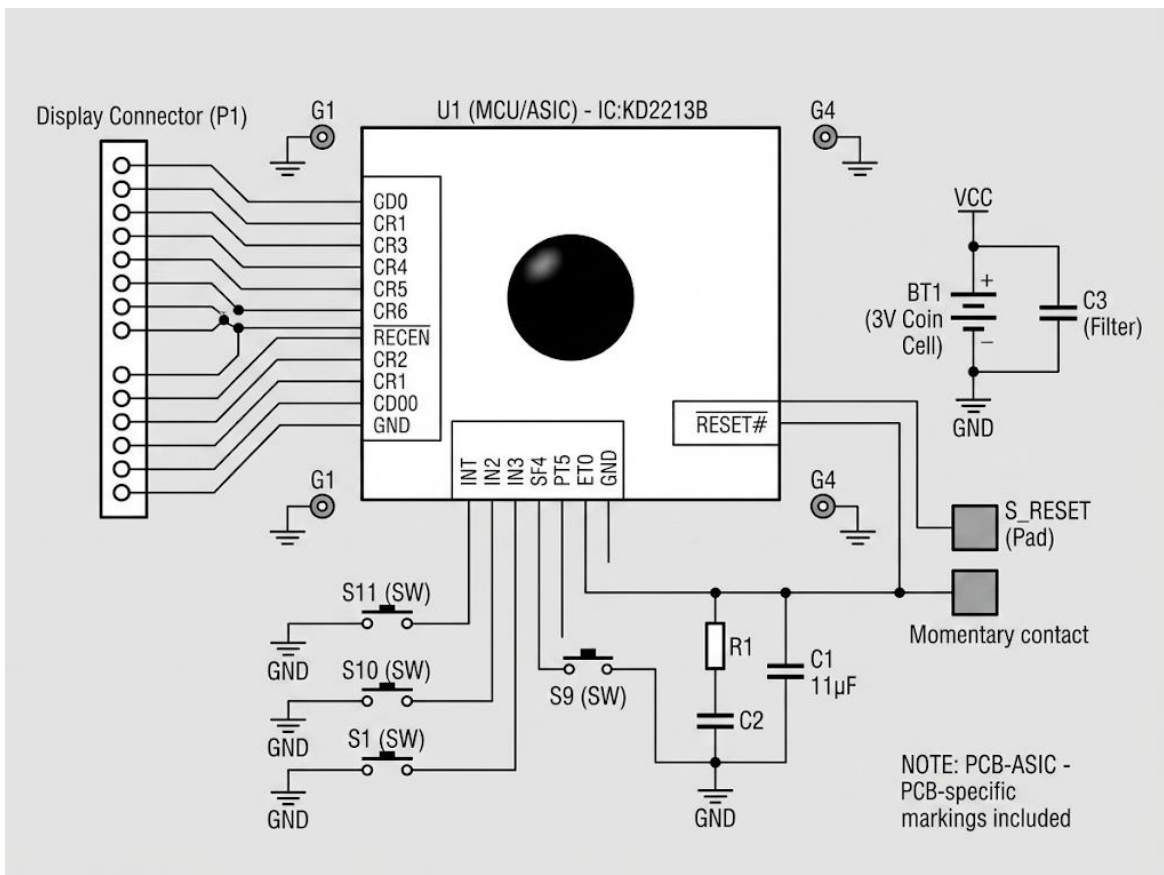
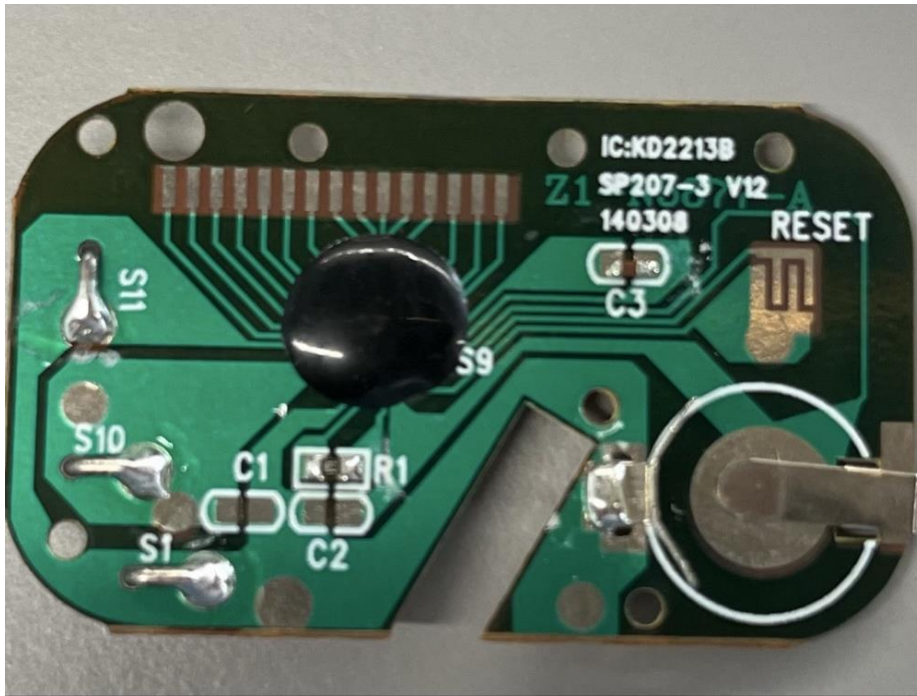


Circuit Diagram Generation Request

User prompt: Create a circuit diagram of this



User prompt: Develop a parts list for this disassembled pedometer.



Response: Enclosure & Structural Components

- **Outer clamshell case:** Silver molded plastic hinged case (top cover and bottom base).
- **Inner PCB bezel/frame:** Silver plastic bezel with cutouts for the LCD and push buttons.
- **Front button fascia:** Silver cover piece labeled "RESET".

Display & Interface

- **LCD panel:** Raw segment LCD glass.
- **Elastomeric connector (Zebra strip):** Pink/black rubberized interconnector strip for LCD-to-board contact.
- **Reset button cap:** Orange molded plastic actuator.
- **Conductive rubber keypad pill:** Dark gray elastomeric membrane contact.

Electronics & Power

- **Printed Circuit Board (PCB):** Green single-sided board with chip-on-board (COB/blob) MCU (IC: KD2213B), integrated battery clip, passive surface-mount components, and trace pads.
- **Coin cell battery:** LR1130 (or equivalent 1.5V alkaline / 3V lithium button cell).

Motion Sensing Mechanism

- **Pendulum arm / Hammer:** Curved metal swing arm with pivot hole and balance grooves.
- **Pendulum spring:** Fine coil/hairspring for pendulum return contact.
- **Pivot pin / bushing:** Small cylindrical black retaining pin.

Fasteners & Hardware

- **Retaining bracket:** Small white 2-hole plastic anchor strip.
 - **Self-tapping housing screws:** 7 small Phillips pan-head screws (fastening the PCB, inner frame, and housing).
-

User prompt: Estimate the manufacturing costs

Response: Based on standard high-volume production runs (50,000 to 100,000+ units) using typical Shenzhen/OEM consumer electronics manufacturing processes, the estimated unit manufacturing cost (COGS) for this mechanical pedometer is **\$0.65 to \$1.15 per unit**.

Subsystem / Category	Component Description	Estimated Cost (USD)
Electronics & Display	Custom ASIC (COB / epoxy blob), single-sided FR4/CEM-1 PCB, passives (R ₁ , C ₁ –C ₃), raw segment LCD, zebra elastomeric connector	\$0.20 – \$0.35
Motion Sensor	Stamped metal pendulum hammer, fine hairspring, pivot pin	\$0.05 – \$0.10
Plastic Enclosure & Buttons	Injection-molded ABS clamshell (top/bottom), inner frame/bezel, button cap, silicone contact pill	\$0.15 – \$0.25
Hardware & Power	LR1130 / AG10 button cell, stamped battery terminal clip, 7× miniature self-tapping screws	\$0.06 – \$0.10
Assembly & Testing	Manual/semi-automated mechanical assembly, screw driving, function testing, and calibration	\$0.10 – \$0.20
Packaging & Documentation	Individual blister pack or printed tuck box, paper user manual, bulk master carton	\$0.09 – \$0.15

Subsystem / Category	Component Description	Estimated Cost (USD)
Total Estimated Unit Cost		\$0.65 – \$1.15

Non-Recurring Engineering (NRE) & Tooling Costs

- **Injection Molds (ABS housing & frames):** \$8,000 – \$15,000 (multi-cavity steel molds for clamshell covers, inner frame, and button caps).
- **Metal Stamping Tooling:** \$1,500 – \$3,000 (progressive dies for the pendulum arm and battery contacts).
- **Custom Segment LCD Mask:** \$800 – \$1,500 (tooling fee for custom segment layouts).
- **PCB Tooling & Stencils:** \$200 – \$500.